

INFLATION

It is the persistent and generalized increase in the price level over a given period of time. Persistent increase in the price level means that the prices continue to increase once they have started to increase. The reason is that an initial increase in the prices will lead to a decrease in the purchasing power or real income of labors who bargain for an increase in the wage rate. This increase in the wage rate causes the cost of production to rise and profits to fall. In order to maintain the profits the producers increase the prices of goods and services which will again decrease the real income of labors and they will demand for higher wages to maintain their purchasing power. This will again increase the cost of production and a second round of increase in the price will begin. This process provides persistency in the price level and is called wage-price spiral

Generalized increase means that prices of all goods and services which have been selected to calculate inflation have to be considered and inflation is the average increase in the prices of all the goods and services. So an increase in the prices of one or few goods is not considered as inflation. Although there may have significant contribution in the current rate of inflation. E.g. an increase in the price of oil can significantly affect the rate of inflation.

The most important cause of inflation is the supply of money in an economy. An increase in the supply of money with given quantities of goods and services will create a situation where too much money chasing too few goods. This will cause the prices to increase and the increased amount of money will still buy the same amount of goods and services i.e. the purchasing power of money will fall. Monetarists believe that sustained inflation would only occur if there was an increase in the money supply.

TYPES OF INFLATION

1. **Hyperinflation**: In this type of inflation, the price level rises very sharply and in its extreme situation people lose confidence in money and stop accepting it as a medium of payments. In this situation people start exchanging goods with the goods and barter starts. It is also referred to as galloping or runaway inflation. This type of inflation is usually associated with war time conditions where most of the resources are employed in the production of war equipment and few resources are left to produce consumer's goods this cause severe shortages of consumer goods causing their prices to rise rapidly. Germany experienced this type of inflation in 1923 and by the end of 1923 the price level increased by one million times.
2. **Suppressed inflation**: In this type of inflation, the price level rises as a result of excess of aggregate demand (AD) over and above the available supplies of goods

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and services. The rate of inflation could be in two digits and this type of inflation usually associated with developing countries.

3. **Mild inflation:** In this type of inflation, the price level rises at a slower rate which is usually under 3%. This type of inflation is usually observed in developed countries where AD and AS conditions are fairly stable.

CAUSES OF INFLATION

DEMAND PULL FACTORS: It is a situation in which the aggregate demand persistently exceeds aggregate supply at current prices. This excess of demand, with a given supply of goods and services, leads to an increase in the prices. When the economy reaches full employment level of output, a further increase in aggregate demand will only lead to an increase in the price level.

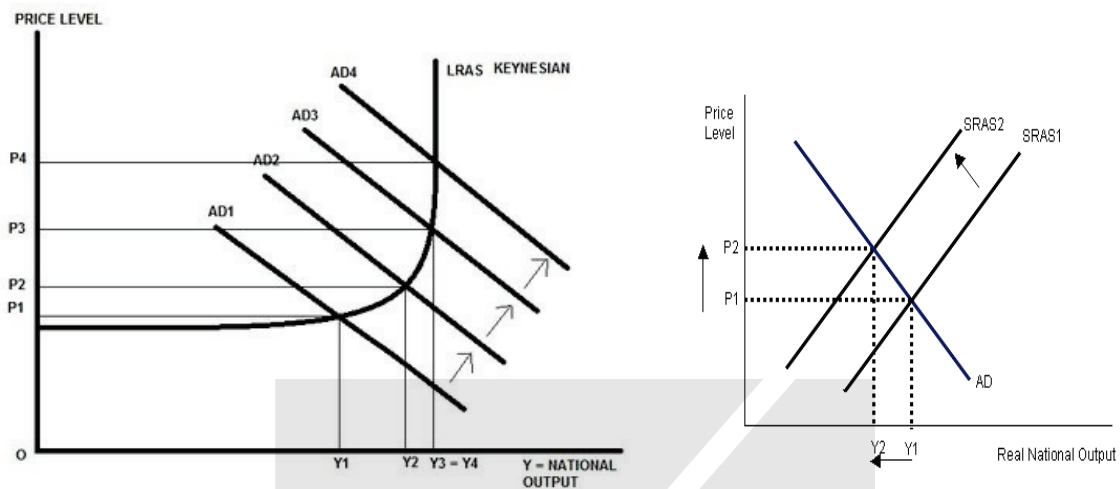
$$AD = C + I + G + (X - M)$$

"C" represents consumption expenditure, "I" represents investment expenditure, "G" represents government expenditure and (X - M) represents expenditure on net exports.

Keynesians assume that the long run aggregate supply (LRAS) curve is horizontal, upward and then vertical. The rising level of demand is pulling the price level up, hence the name demand pull inflation. There are many situations where the economy at full employment faces excess of AD over the available supplies of goods and services.

During the war most of the resources are employed which leads to an increase in income and an increase in demand. On the other hand, the supply of consumer goods and services falls greatly as most of the resources are diverted in the production of war equipment, this excess of aggregate demand over aggregate supply leads to sharp increase in the price level. Demand pull inflation may also occur when a country tries to achieve faster rates of economic growth. To achieve growth the resources of the economy are diverted towards the production of capital goods which leads to shortages of consumer goods thereby increasing their prices.

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Demand inflation occurs when, at full employment, for e.g. the government expenditure increase and is financed through borrowing from central bank. The government expenditure increases the income level in the economy causing the demand for goods and services to increase. With the given supplies of goods and services, this increase in aggregate demand will pull up the price level.

1. **COST PUSH FACTORS:** There are factors which tend to increase the cost of production causing the profits of the producers to reduce. As a result the producers react by increasing the prices of goods and services to shift part or whole of the increased cost of production on the consumers.
Cost push inflation also occur when the price of an imported raw material increase which significantly increase the cost of production. This can happen as a result of an increase in the world prices of raw material or because of depreciation of domestic currency which has made the imported raw material more expensive in terms of domestic currency. The oil price shocks of the mid 70s and early 80s are good example of large increase in costs causing inflation. The militant trade unions made things worse by insisting on above inflation pay rises to make up for the price rises plus any unpredictable future rises in the price level. If input prices increase firms will either have to increase their prices or reduce their profit margins.

Devaluation in the exchange rate is also likely to cause inflation; firstly this will cause an increase in the prices of imports which will be passed on to consumers, considering demand for imported goods to be inelastic, a raw material. Secondly

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there will be an increase in exports revenue because of exports appearing cheaper to foreign customers, this will result in higher aggregate demand.

An increase in the wage rate without increase in the productivity of labor will lead to cost push inflation. E.g. government intervention of minimum wage legislation will increase the cost of producing a given level of national output leading to a fall in aggregate supply. The producers will pass the burden of increased cost on the consumers in the form of increased prices of goods and services.

An increase in indirect tax is also an important cause of cost push inflation. Indirect tax such as value added tax (VAT) is imposed on goods and services which increases the cost of production and producers pass some or entire tax on consumers in the form of increased prices. Even goods imported into a country can cause inflation i.e. imported inflation, this occurs when there is an increase in the price of goods which is being imported, the effect in terms of increase in cost will occur when demand for those goods will be inelastic for e.g. a raw material.

The link between demand-pull inflation and cost push inflation

Some changes will both increase aggregate demand and increase costs of production. For example, a fall in a country's foreign exchange rate may both raise the price of imported raw materials (cost-push inflation) and increase export revenue (demand-pull inflation).

Also, once inflation occurs, there is a risk that an inflationary spiral may occur. Demand-pull and cost-push factors may interact and reinforce each other and an inflationary spiral may develop. Higher government spending on state pensions, for instance, may increase aggregate demand. The resulting higher prices may encourage workers to call for higher wages. If they do gain higher wages, costs of production may increase, causing aggregate supply to decrease. The higher wages may increase consumer expenditure and so the upward pressure on prices would continue.

2. **MONETARY INFLATION:** This situation occurs when there is an increase in money supply or reduction in the interest rate

When there is an increase in the money supply, this will result in a fall in interest rate. A fall in interest rate will increase the consumption rather than saving and along with this there will be increased amount of borrowing and therefore more investment. Along with this a fall in interest rate will reduce exchange rate as they have a positive relationship. Depreciation of our currency will make exports appear cheaper and imported goods to appear expensive, which is likely to cause balance of

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trade surplus. All these actions will increase aggregate demand and at full employment level of output there will be inflation.

The consequences of inflation

Inflation has both possible costs and potential benefits.

The possible costs of inflation

- **A reduction in net exports:** Inflation may reduce the international competitiveness of a country's products and so increase import expenditure and lower export revenue. This may result in balance of payments problems.
- **An unplanned redistribution of income:** Some people may gain and some people may lose as a result of inflation. For instance, if the rate of interest does not rise in line with inflation, borrowers will gain and lenders will lose. This is because borrowers will pay back less in real terms and lenders will receive less. Savers will also lose out as the real value of their savings will fall.
- **Menu costs** are the costs involved in changing prices. Menu costs affect firms, for example catalogues, price tags, bar codes and advertisements have to be changed. Changing prices involves staff time and is unpopular with customers.
- **Shoe leather costs** are the costs involved in moving money from one financial institution to another in search of the highest rate of interest.
- **Fiscal drag** (also known as 'bracket creep') occurs when the income levels corresponding to different tax rates are not adjusted in line with inflation. As a result, people and firms are 'dragged' (pulled) into higher tax brackets. However, it can be argued that this is a cost of an inefficient tax system rather than a cost of inflation.
- **Discouragement of investment:** Unanticipated inflation can create uncertainty and so make it more difficult for firms to plan ahead. This may discourage firms from investing, which will have an adverse effect on economic growth.
- **Inflationary noise** (also called 'money illusion') arises when inflation causes consumers and firms to confuse price signals. Inflation can make it difficult to assess what is happening to relative prices. A rise in the price of a product may not mean that it has become more expensive relative to other products; the product may have risen in price by less than inflation and so may have become cheaper in real terms. Inflationary noise can result in consumers and firms making the wrong decisions. For example, firms seeing the price of their products rising may increase output when the higher price is the result of inflation rather than increased demand for their products. This may result in a misallocation of resources.

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- **Inflation causing inflation:** Inflation may generate further inflation as consumers, workers and firms will come to expect prices to rise. As a result, consumers, workers and firms may act in a way that will cause inflation. For example, workers may call for higher wages, firms may raise prices to cover expected higher costs and consumers may seek to purchase products now before prices rise further. Inflationary expectations can be powerful and it can be difficult for a government to change such expectations.

The potential benefits of inflation

- **Stimulates output:** A low and stable inflation rate caused by increasing demand may make firms feel optimistic about the future. In addition, if prices rise by more than costs, profits will increase, which will provide funds for investment. Investment may also be encouraged as the real rate of interest may fall and consumer expenditure may rise as higher money incomes will make people feel better off even if, in real terms, they are not.
- **Reduces the burden of debt:** Real interest rates may fall due to inflation or may even become negative. This is because money interest rates do not tend to rise in line with inflation. As a result, debt burdens may fall. For example, those who have borrowed money to buy a house may experience a fall in their mortgage payments in real terms. A reduction in the debt burden may stimulate consumer expenditure which, in turn, could lead to higher output and employment.
- **Prevents some unemployment:** Firms in difficulties may have to reduce their costs to survive. For many firms, wages form a significant proportion of their total costs. Inflation would enable them to reduce the real costs of labour by either keeping money wages constant or by not raising them in line with inflation. During inflation, workers with strong bargaining power are more likely to be able to resist cuts in their real wages than workers who lack bargaining power.

Factors affecting the consequences of inflation

The effects of inflation depend on:

- **The cause of inflation.** Demand-pull inflation is likely to be less harmful than cost-push inflation. This is because demand-pull inflation is associated with rising output whereas cost-push inflation is associated with falling output.
- **The rate of inflation.** A high rate of inflation is likely to cause more damage than a low rate especially if the high rate develops into hyperinflation. Indeed, hyperinflation can lead to households and firms losing faith in the currency and may bring down a government.

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- **Whether the rate of inflation is accelerating or stable.** An accelerating inflation rate, and even a fluctuating (constantly changing) inflation rate, will cause uncertainty and may discourage firms from investing. The need to devote more time and effort to estimating future inflation rates will increase costs.
- **Whether the inflation rate is the one that has been expected.** Unanticipated inflation, which occurs when the inflation rate was different from that expected, can also create uncertainty and so can discourage some consumer expenditure and investment. In contrast, if households, firms and the government have correctly anticipated inflation, they can take measures to adapt to it and so avoid some of its potentially harmful effects. For instance, firms may have adjusted their prices, money interest rates may have been changed to maintain real interest rates and the government may have adjusted tax brackets, raised pensions and public sector wages in line with inflation.
- **How the inflation rate compares with the rate of other countries.** It is possible for a country to have a relatively high rate of inflation, but if it is below that of competing countries its products may become more internationally competitive.

RETAIL PRICE INDEX (RPI)

The value of a good generally means the amount of other goods which will be given in exchange for it. In practice, the relative values of different goods are usually compared directly by pricing the goods, i.e. by expressing their value in terms of common commodity, money.

When, for instance, the prices of goods in general have risen, it means that fewer goods will be bought by the same amount of money. In other words that value of money has fallen. Similarly, when all the prices have fallen, the value of money has risen and thus consumers' purchasing power has increased.

The value of money is thus its value in terms of goods and services purchased. Its value is seen indirectly through the prices of other things. It also means to say that the study of prices is the study of changes in the value of money.

In order to measure changes in the value of money, we have to devise a system whereby variations in the prices of things in general can be recorded. Generally we measure the value of money in terms of the retail prices of goods bought by the consumers.

CONSTRUCTION OF RPI: It is the index which is normally used to measure the rate of inflation and is a weighted index of prices of goods and services purchased by an average

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household. Movements in this index therefore show that the cost of living has changed. Annual percentage increases in the RPI shows the rate of inflation. Thus if the RPI went up from 100 to 110 during a year, we could say that rate of inflation was 10%.

Calculating a price index is a complicated process. In the UK, each monthly surveyor are sent out to record for 600 items which have been selected to calculate RPI because these are the goods which are normally bought by the people in their monthly expenditures and is called the representative basket of goods and services. Prices are recorded in different areas of the country as well as in different types of retail outlets such as corner shops and supermarkets. These results are averaged out to find the average prices of goods and services and the figure are converted into index number form. Changes in the prices of food are more important than changes in the prices of tobacco. This is because a larger proportion of total households' income is spent on food than on tobacco. Therefore, the goods are assigned numbers or weight which represents the proportion of income spends on each commodity, so it shows the importance of the good in the representative basket. As a result, the RPI weighs rather heavily the prices of commodities on which consumers spend much of their income, and weighs rather lightly the prices of commodities on which consumers spend a little of their income.

The index is taken as 100 in the base year which changes in the subsequent years according to the proportionate changes in prices of the commodities. These index number are then multiplied by their weights and are added together to calculate the weighted index. This index is then compared with the base year's index to find RPI.

GOODS	WEIGHT	INDEX	$W \times I$
A	0.5	120	60
B	0.33	150	49.5
C	0.17	200	34
			143.5

Rate of inflation in year 1 is therefore 43.5% (143.5 - 100)

Calculating a specific inflation rate: Normally, we want to know how much prices have increased since last year, or since we bought our house, or perhaps how much prices will increase by the time we retire or our kids go to college.

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Fortunately, the method of calculating inflation is the same, no matter what time period we desire. We just substitute a different value for the first one. So if we want to know how much prices have increased over the last 12 months (the commonly published inflation rate number) we would subtract last year's index from the current index and divide by last year's number and multiply the result by 100 and add a % sign.

The formula for calculating the inflation rate looks like this:

$$((B - A)/A) * 100$$

So if exactly one year ago the consumer price index was 178 and today is 185, the calculations would look like this:

$$((185 - 178)/178) \times 100$$

3.93% inflation rate over the sample year.

PROBLEMS/LIMITATIONS of calculating RPI:

One of the problems encountered in calculating RPI is the choice of base year. It is particularly important because a wrong choice will distort the situation. The base year should be a year when the prices are reasonably stable. It should be a period of free from abnormal conditions like war and famine.

Another problem encountered in the compilation of RPI is to decide how much of each commodity to select. This is the problem of weighting because different weights give different results. The choice of weights is a big problem because it requires accurate information and the data regarding the expenditure pattern of the households. However, many people resent inquiries in to the way in which they spend their money, and even when they are prepared to give this information, they are often unwilling to admit the full amount they spend on things such as tobacco, alcohol and entertainment.

As standard of living of people rises, their distribution of expenditure also changes. As the standard of living raises, expenditures on luxuries as a percentage of their income increases and expenditure on basic necessities such as food, shelter and clothing as a percentage of income tends to fall. Hence weight attached to these items must be changes in accordance to changes in the expenditure pattern.

Another problem arises as price index cannot be used over a long period of time. This is due to the facts that

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1. New goods may be introduced
2. Demand for goods may change according to changes in taste and fashion
3. The age distribution of population may also change
4. In countries where there is a large rural sector, the use of simple price index for the whole country is misleading.

Another problem encountered in calculating RPI is the right choice of goods for the basket. The assumption that everybody spends their income on the same range of goods and buys similar proportion of them is false. Generally, we say that the higher the standard of living, the greater these differences will be. Thus, the price index can only be used to show variations in the cost of living only for a particular income group on whose expenditure it is based.

USES OF RPI

The index of retail prices is important because of the implication of such changes on the cost of living of people and on wages, when the prices of goods and services increase while the money wages remain constant, the purchasing power of money will fall i.e. people will be able to buy lesser amount of goods and services with the same amount of money. An increase in the prices therefore is likely to be followed by demands for higher wages and salaries.

RPI is of great importance to many workers and their unions since an increase in RPI is frequently felt to provide a justification for the demand for higher wages because workers and their unions are concerned with real wages (wages after adjusting for inflation) and not just money wages.

The RPI is of great academic interest because it enables us to know about the changes in the standard of living of people in an economy.

RPI is also useful for the government because it helps in the formulation of some of the policies e.g. the rise in the wage rate required to compensate the labors, changes in the rate of interest or the money supply required to reduce the rate of inflation to some target level and so on.

Savers also like to know what will be the future value of their savings because the high inflation rate erodes away purchasing power of money and savings will be discouraged.

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ANTI INFLATIONARY POLICY

The policy measures adopted will depend on the cause of inflation. These policies can be directed towards the control of either AD or AS, and are considered as demand side and supply side policies.

DEMAND SIDE POLICIES

Aggregate demand can be controlled by using contractionary fiscal and monetary policies.

1. **Fiscal policy:** It is the policy of the government in which taxes and government expenditure are manipulated to achieve certain aims in the economy e.g. controlling inflation.

When the economy is experiencing demand pull inflation at full employment level of output, the government uses contraction or deflationary fiscal policy to reduce AD so that demand pull inflation can be controlled.

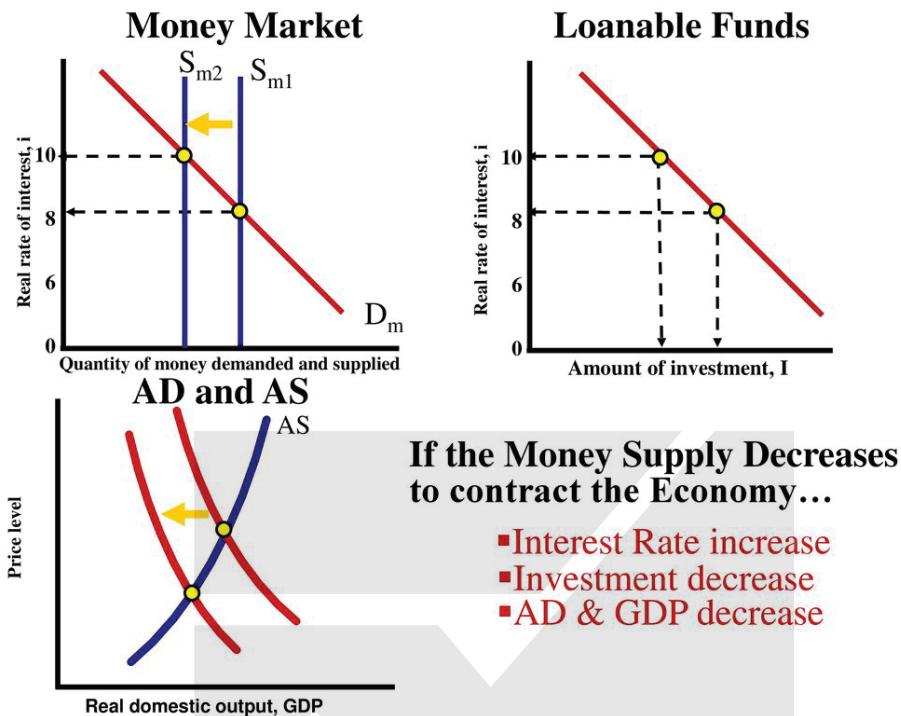
This is done by increasing taxation and reducing government expenditures. An increase in tax on the incomes of people will lead to lower after tax incomes causing consumption demand to reduce. Along with that a rise in corporation tax will reduce retained profits thus reducing investment.

Similarly reducing government expenditures will directly reduce the demand for all those goods which were demanded by the government. Both these measures will reduce the aggregate demand and demand pull inflation will be controlled.

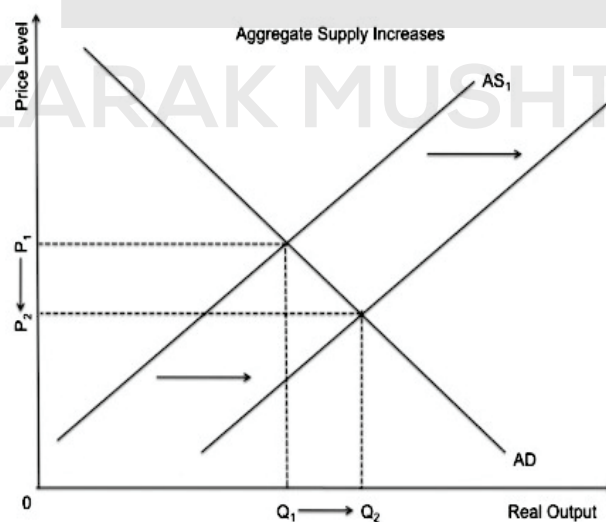
2. **Monetary policy:** It is the policy of the government in which the government manipulates the money supply and the rate of interest to achieve macro-economic objectives such as maintaining inflation at low levels. One of the causes of demand pull inflation is the increase in the money supply with proportionally lesser or no increase in the output of goods and services. To combat demand pull inflation in this situation the supply of money is reduced so that it matches the growth of output of goods and services.

To achieve a reduction in the growth of money supply the government may increase the rate of interest making it expensive for the consumers and producers to borrow from the banks at higher interest rates. This will reduce consumption and investment demand. A rise in interest rate will appreciate the value of country's currency, this will make exported goods appear expensive and imported goods appear relatively cheaper, resulting in trade deficit, causing a fall in incomes. All these actions reduce aggregate demand causing demand pull inflation to reduce as well.

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SUPPLY SIDE POLICIES: These are the policies which are aimed at reducing the costs of production so that aggregate supply could be increased and the cost push inflation could be controlled. The government may provide grants for firms (domestic as well as foreign multinationals) setting up in under- developed areas of country. They may also reduce tax on the profits of the firms so that the after tax profit increases and the firms are encouraged to produce more causing aggregate supply to increase and the cost push inflation could be controlled.



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Government can reduce tariffs on the imported raw material; the cost of production in the domestic currency will be reducing which helps in reducing the cost push inflation.

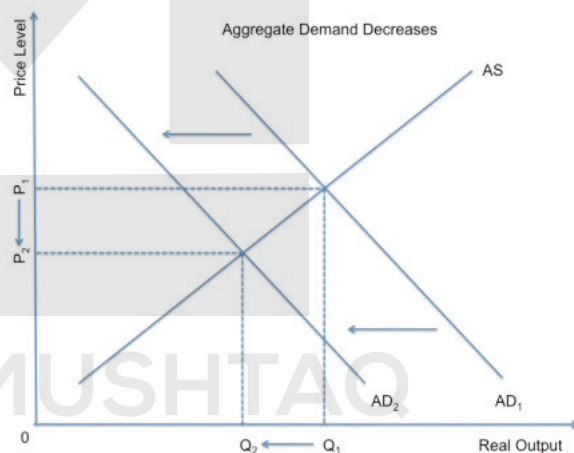
Government can provide education and training to labor force, their productivity increases.

Provision of subsidies to the producers also helps in reducing the costs of production having a dampening effect on the cost push inflation. Supply side policy is the only policy which can achieve three macroeconomic objectives without conflicting i.e. low rate of inflation, economic growth (increase in real output) and high rate of employment.

Deflation

Price deflation is when the rate of inflation becomes negative, i.e. the general price level is falling and the value of money is increasing. Some countries have experienced deflation in recent years - good examples include Japan and China. In Japan, the root cause of deflation was very slow economic growth and a high level of spare (excess) capacity in many industries that was driving prices lower. In China, economic growth has been rapid - but the huge amount of capital investment and rising productivity has led to economies of scale being exploited and a fall in production costs.

A sustained fall in the general price level in an economy, or deflation, is also a cause for concern. This is because deflation will usually occur when the demand for goods and services is falling. Firms are able to sell fewer goods and services and so cut their prices and lose profits. Because they do not need to produce so much, and as they try to cut their costs, firms reduce the size of workforces and may even require their remaining workers to take a cut in wages.



Household's incomes fall as unemployment rises, furthering reducing demand for goods and services. The value of debts held by people's and firms will also rise in real terms as prices fall and the burden of making loaned repayments will increase. Eventually the economy goes into deep recession as demand, output, the demand for labor, and incomes continue to fall.

Many firms may go out of business because demand continues to fall and they are unable to make any profits no matter how much they cut their prices; deflation will result in downward multiplier effect i.e. income, output and employment will decrease.